The Alberta Buck - Demurrage and the Jubilee Fund

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Abstract

Held and issued-but-unspent BUCK pay demurrage on the BUCK remaining in the account. Each BuckCredit position earns Jubilee relief on the BUCK it actually issued, proportional to issuance and elapsed time. The total gross demurrage basis is all outstanding BUCK, including BUCK held by Jubilee. The fund receives the charges and may invest its assets; it pays the relief when issuance is retired. Gross demurrage includes charges paid by Jubilee to itself, so cumulative gross demurrage and BUCK actually held by the fund are distinct.

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1 Status

This is the intended monetary design, clarified through 2026-10-29. The executable Python reference model and its tests exercise the mathematics below. The current Solidity contract and Rust mirror still use a linear charge on a holder's gross raw balance and advance Jubilee on issuance; they do not yet implement the model in this document. Their existing tests establish the old behavior, not conformance with the new one. The detailed implementation audit is Net-balance demurrage: audit and redesign. The proposed division of responsibility among Buck, each credit NFT, Jubilee and each insurer's pool is in Buck, credit facilities, Jubilee, and insurer pools.

Earlier versions of this document described linear demurrage on the holder's raw balance and contained older storage sketches. Those formulas are superseded; the previous text remains available in the Git history.

2 Issuance, holdings, and relief

A BuckCredit authorizes credit; BUCK actually issued into existence form the demurrage base and earn relief. When a facility issues BUCK into an account's issued-but-unspent balance, they are already issued and pay demurrage even before they are transferred. Merely potential capacity that has not been issued is not BUCK. While issuance remains outstanding, the corresponding net BUCK are in issued-but-unspent balances, ordinary accounts, pool or other service accounts, or Jubilee. The underlying asset's insured coverage and a user's not-yet-issued capacity are different quantities from outstanding issued BUCK.

Let `I(t)` be total outstanding BUCK issuance, `r` the nominal annual rate, and `P_j(t)` the outstanding issuance attributable to credit position `j`. As issuance changes, the system checkpoints the old base and starts integrating the new base. Over an interval,

```
BUCK-years issued       = integral(I(t) dt)
gross demurrage         = r * integral(I(t) dt)
position j BUCK-years   = integral(P_j(t) dt)
position j target relief = r * integral(P_j(t) dt)
```

BUCK-years are the integral of outstanding issued BUCK, including issued but unspent BUCK, not the age of a potential credit limit. When an issuance position is reduced, its target relief is computed from its BUCK-years. Jubilee first redeems a proportion of its investment positions and computes the actual relief from its available full-value BUCK; the account repays the remaining principal. Actual relief can be lower or higher than the target, but is bounded by the principal being reduced and by the available net BUCK. In the reference, a partial reduction takes a pro-rata share of that position's accumulated BUCK-years. Separate positions retain separate issuance histories. For issuance held for 50 years at a nominal 2% per year, the target reaches the entire position; the actual payment depends on the fund and cannot exceed the position. Gross issuance-seconds continue accumulating for system demurrage after the target reaches that cap. Claimable position seconds are capped before new principal is added, so excess age beyond 50 years cannot immediately relieve newly issued BUCK. The reference keeps one average age within each NFT; after new issuance, the combined principal accrues toward its combined cap.

An account's received/held BUCK, issued-but-unspent BUCK, and the outstanding issuance of each NFT it owns are not interchangeable. Its ERC-20 `balanceOf` reports the full spendable balance: held plus issued-but-unspent net BUCK for an ordinary account. Separate views can expose each amount. Spending issued BUCK does not issue it again, and receiving BUCK does

not automatically retire an NFT. Relief reduces a liability only when it is paid. The target and the actual relief computed after liquidation must each be visible; a target shortfall is not an additional guaranteed claim.

3 Demurrage on remaining balances

Demurrage already transferred to Jubilee no longer belongs to the former holder and is not charged again to that holder. Held and issued-but-unspent BUCK follow the same net-balance rule. For an idle account with a remaining balance `H` and no other flows, the Python reference uses a nominal continuous rate:

```
dH/dt = -r H
H(t)  = H(0) * exp(-r t)
```

The reference uses continuous compounding because the contract has a per-second nominal rate and a result must not depend on the frequency of permissionless settlement. If a future implementation instead chooses an effective annual rate or another schedule, that convention and its rounding must be specified and used consistently in contracts, models, and tools.

New incoming BUCK join the remaining balance at their arrival time. Outgoing BUCK leave the balance at their outgoing time. Subsequent charges apply to the resulting net holdings, never to fees already transferred out. For example, an account with 0.5 BUCK remaining incurs approximately 0.009901 BUCK of continuous demurrage over the next year at 2%; it does not pay 2% of a much larger historical raw balance.

The fund's BUCK also experience demurrage. If the fund holds BUCK directly, its demurrage is transferred to the fund itself: it counts in total gross demurrage, but the self-transfer has no net effect on fund assets. A fund position in a basket or desk requires separate accounting for the underlying BUCK, beneficial ownership, realizable asset value, and its returns. It cannot assume that a mark or a cumulative demurrage figure is immediately spendable BUCK.

4 The 100 BUCK, 50-year example

One credit issues 100 BUCK into an ordinary account's unspent balance. Its ordinary held balance starts at zero and `balanceOf` reports the full spendable unspent amount. Nothing is transferred, invested, retired, or added for 50 years. At a nominal continuous 2% rate, the Python reference produces, ignoring rounding:

Years	Unspent / balanceOf	Jubilee BUCK	Gross demurrage	Jubilee self-demurrage	Target relief
0	100.000000	0.000000	0.000000	0.000000	0.000000
1	98.019867	1.980133	2.000000	0.019867	2.000000
25	60.653066	39.346934	50.000000	10.653066	50.000000
49	37.531110	62.468890	98.000000	35.531110	98.000000
50	36.787944	63.212056	100.000000	36.787944	100.000000

The 100 BUCK gross demurrage is the sum of 63.212056 transferred from the unspent balance and 36.787944 paid by Jubilee to itself. The latter is counted as a charge, but does not create an additional 36.787944 BUCK in the fund. Unspent balance plus fund still hold 100 BUCK, matching

outstanding issuance. Crediting Jubilee with all 100 **in addition** to leaving 36.787944 unspent would create 36.787944 BUCK without a source.

The target relief at year 50 is 100, but the fund directly holds only 63.212056 in this no-return, no-transfers example. It needs another 36.787944 BUCK to pay the full target. Acquiring that amount from the owner, for example by selling other assets or realizing returns, would leave 100 in the fund; a 100 BUCK Jubilee payment can then extinguish the credit. The external consideration and market price are not represented by the closed BUCK ledger. Under the available-relief rule the actual relief is computed from the fund's net liquid BUCK after any proportional redemption; the borrower pays the remainder. No separate fixed 100 BUCK payout is assumed.

5 Fund investments and liquidity

Jubilee may put its assets to work in the BUCK-maximizing BuckBasket and, when available, desk implementations that seek other assets such as USDC. A useful accounting implementation must distinguish:

- cumulative gross demurrage, including Jubilee's self-payments;
- liquid BUCK owned by the fund;
- deployed positions, their acquisition cost, current mark and realizable proceeds after liquidity, price impact and transaction costs;
- other assets and the BUCK they can actually acquire;
- each credit position's accrued and payable relief; and
- the cash and positions reserved for anticipated relief payments.

For the fund's cash ledger, separate cumulative 256-bit counters can record full-value demurrage received, relief paid, BUCK invested, and net BUCK withdrawn. With no other flows, liquid net BUCK equals received demurrage plus investment withdrawals, less relief payments and investments. Other inflows and outflows are recorded separately. Demurrage paid by the fund to itself cannot be added to net incoming cash a second time. A running average cost basis on the pooled investment additionally identifies gains or losses only as the corresponding shares are withdrawn. The net of either pair of cumulative counters can be negative after gains; unsigned totals must not be subtracted without signed or otherwise guarded arithmetic.

When a BuckCredit position is reduced, the fund computes relief from the selected fraction of its net cash and marked basket receipt, without selling the receipt during that transaction. The calculated net relief must be no more than 10% of the available net cash, otherwise the reduction reverts. The borrower supplies the difference; if the borrower lacks that amount, the reduction cannot complete. A separate keeper can redeem basket shares to replenish the cash reserve. A modeled investment return must have a source, such as a transfer from a counterparty, a realized external asset sale, or new, separately recorded issuance. It must not appear solely because the fund paid demurrage to itself. The reference includes an idealized BUCK-only strategy that pays demurrage into fund cash and liquidates at its current mark, plus returns transferred explicitly from a counterparty. The strategy's mark falls as its BUCK pay fees, while the same fee increases fund cash: counting the mark decrease alone as a loss of total fund value would be misleading. The reference does not model actual basket prices, collateral values, transaction costs, delay, or market depth.

An earlier available-relief experiment scales the target by post-redemption net cash divided by a 10% reserve target, capped at net cash and principal. That signal can mistake uninvested cash or redemption proceeds for returns. In a two-facility idle example it lets the first reducer receive the entire fund, leaving the second unable to close. It is retained only as a comparison, not a safe contract rule.

A second earlier rule allocated the fraction `f` of pre-sale cash plus actual proceeds from selling fraction `f` of the position, capped by principal. It remains an explicit reference comparison, not the current contract policy. The marked rule allocates the same fraction of net cash plus the receipt's marked value; if an actual sale realizes exactly the mark with no costs, the two rules agree. A mark is not a guarantee of liquidity or realized price. See the marked-relief analysis for the comparison and remaining audit work.

The simulator retains fixed-2%, reserve-ratio and proportional-sale rules as comparison diagnostics. Its command-line retirement defaults to the marked rule. Credits of different ages make their share of outstanding relief very different from their share of issued BUCK. The reference sums its positions to obtain the benchmark; the experimental claim book maintains the capped aggregate incrementally, with bounded permissionless processing when facilities reach 50 years.

The experimental replacement does not sell the fund's basket position inside a reduction. It allocates the selected fraction of the fund's **net liquid BUCK plus the net marked value of its basket receipt**, and reverts if that relief exceeds 10% of liquid net BUCK. The selected fraction is the reduced facility's capped BUCK-seconds divided by all capped outstanding BUCK-seconds, including any matured principal assumed by Jubilee and not yet burned. Consequently the effective rate is the nominal 2% multiplied by the whole-fund mark divided by the aggregate relief benchmark, subject to the principal cap. The isolated Solidity reduction and renewal paths implement this rule; their oracle, mark-to-market and executable-liquidity assumptions still require further review and simulation. See the marked-relief analysis for a numerical example, the 50-year aggregate and valuation limitations.

At 50 years of a facility's average issuance age, its owner can instead release the lien without providing BUCK. The prototype burns the NFT and records the unburned principal as Jubilee's obligation, retaining it in outstanding supply and the benchmark until the fund later burns it. A premium deposit remains owed by the original insurer pool in nominal BUCK; if the pool cannot immediately return it, the refund can be retried after lien release. Deferred burns are limited to 10% of current net fund cash per operation. A strict limit can leave a small residual obligation if fewer than ten raw BUCK units are liquid.

If an owner leaves an already mature facility open, its claim does not grow beyond its principal, but the BUCK it issued remains in circulation and continues to pay demurrage. For example, with 100 BUCK issued and held in an ordinary account, after 50 years that account still has approximately 36.79 BUCK and Jubilee has approximately 63.21 net BUCK in the no-investment example. The 100-BUCK target does not imply that 100 BUCK are already liquid in Jubilee. At release the fund must burn what it can under the liquidity rule and retain the rest as an obligation until BUCK are acquired or accrue. Value exceeding a facility's principal remains available to other claims; it may result from later demurrage as well as investment returns.

In both examples the **gross** demurrage assessed over those 50 years is the full 100 BUCK. The difference is that, without an investment return, the approximately 36.79 BUCK of fees paid by the fund to itself do not add net value to its own account a second time. A separate closed-ledger comparison models the entire fund earning 2% annualized: the return is transferred from another holder's existing BUCK, so Jubilee accumulates 100 net BUCK at year 50 and total supply remains 100. No new credit is necessary, though the return has a real economic source and is not guaranteed. The fund in that comparison is fully invested; a liquid reserve does not itself earn that

investment return. Run `make nix-venv-demurrage-returns` to reproduce the comparison, or add `DEMURRAGE_ARGS"–json"=` for its rows.

6 Account state and carrying services

The present contract uses one signed raw account balance and a timestamped `BUCK-seconds` accumulator. Its positive side represents gross held BUCK and currently uses the accumulator for the holder's accrued fee; its negative side represents actual credit drawn and uses the accumulator to integrate the outstanding lien for relief. Zero must have zero accumulated seconds. A sign change must settle the old interpretation before recording the new one. This is a description of the current code, not the proposed model: the current account sign cannot represent held BUCK, issued-but-unspent BUCK and multiple independent NFT liabilities at once. The proposed design stores the BUCK and credit-facility responsibilities separately and attributes each NFT's relief to that NFT. The signed view of an ordinary positive holding must never turn negative merely because its accumulated fee exceeds its raw amount.

The existing Carrying mode lets service contracts, including AMM pools, report nominal raw balances to preserve ERC-20 reserve accounting, while passing a proportional accumulated fee with a transfer. These nominal amounts are not necessarily the same as unencumbered economic value. The corrected model must preserve a clear distinction between nominal pool balances, remaining value, transferred fee attribution, and the fund's actual assets. For Jubilee's own Carrying balance, a relief payment must deliver its full net value. If the fund has 100 nominal BUCK with 20 BUCK of carried fee, its net balance is 80. To deliver 40 net BUCK under proportional continuous fee allocation, it sends 50 nominal BUCK carrying 10 of fee; 50 nominal and 10 of fee remain, or 40 net. The fund's already-credited self-demurrage must not be credited again on payout. The actual on-chain rounding must follow the chosen accumulator representation; the Python integer example is not yet a derivation for Solidity's buck-seconds rounding. The current sign encoding and Carrying rules must be replaced or adapted; they do not justify charging an already transferred fee again.

Each upstream insurer has its own Carrying premium pool. An initial deposit of 10 full-value BUCK is refunded as 10 nominal BUCK, with that pool's average carried fee. This differs from Jubilee relief, which must deliver full net value. A pool can invest and redeem its positions before paying; an insurer's deposit and refund must not be mixed with another insurer's pool. A renewal can settle an NFT's actual relief, update insurance terms, and optionally add issuance atomically without first releasing its old coverage.

Delegated demurrage also needs an explicit net-balance treatment: its payer must have enough actual remaining value, and the outcome must remain conserved when a payer is depleted or multiple sponsees settle in different orders. See the implementation audit for the existing lazy behavior.

7 Run and verify

```
make nix-venv-demurrage-reference
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--issued 1000000 --years 1"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--years 50 --json"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--years 50 --acquire-all"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--deploy-fund --strategy-return 10"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--deploy-fund --wipe-strategy --retire 50 --re
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--retire 50"
make nix-venv-demurrage-reference DEMURRAGE_ARGS="--retire 50 --payout-policy reserve"
```

```
make nix-venv-demurrage-test
```

The reference is an event-driven, floating-point monetary model, not an EVM or fixed-point implementation. Its tests cover 1 to 1,000-year idle holdings, changes to outstanding issuance, proportional reductions, separate NFT positions, full spendable balances, insurer-pool refunds and atomic renewals, small net balances, settlement-frequency independence, fund self-demurrage, strategy returns and liquidation, zero proceeds, grossed-up Carrying relief, conservation, and insufficient debtor funds. Before contract deployment, implement and cross-check fixed-point precision and rounding, supply and claim conservation, settlement and retirement transitions, pool and delegated-fee semantics, fund liquidity, and migration of any existing account history. The five-year economy and sandboxes must subsequently use the implemented rules; existing runs use the old contract and do not validate this design.